Global Life Expectancy Prediction Analysis

Project Overview

This project develops a machine learning model to analyze the key health, economic, and social factors that influence life expectancy across countries. It examines how indicators such as healthcare access, immunization, education, and income levels affect longevity worldwide.

The goal is to build a predictive system that can estimate life expectancy based on these factors and identify the most important drivers behind global differences.

Dataset Profile

The analysis uses a real-world **World Health Organization (WHO)** dataset containing **2,938 records and 22 features**. It includes country-level health, demographic, and economic indicators.

This dataset is used for supervised learning to predict life expectancy and understand the impact of global development factors.

Exploratory Data Analysis (EDA)

Statistical Summary

- Life expectancy ranges from approximately **36 to 89 years**
- Average life expectancy is around **69 years**
- Adult mortality and GDP show high variability
- Some features contain missing values (e.g., GDP, Hepatitis B, Population)

Data Quality Issues

Missing values were found in several important features:

- Population: 652 missing values

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- Hepatitis B: 553 missing values
- GDP: 448 missing values

These were handled during preprocessing using imputation techniques.

Life Expectancy Distribution

Life expectancy is slightly left-skewed, with most values concentrated between **70 and 80 years**. This indicates that while many countries have high life expectancy, some regions still experience significantly lower values.

Correlation Analysis

- Positive correlation: **Schooling, Income, GDP**
- Negative correlation: **Adult Mortality, HIV/AIDS**

This shows that better education and economic conditions improve life expectancy, while disease burden and mortality reduce it.

Outlier Detection

Outliers were detected using the IQR method. The most affected features include:

- Measles
- HIV/AIDS
- GDP
- Infant deaths

These outliers indicate large differences between countries in health and economic conditions.

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Data Preprocessing

The dataset was cleaned and prepared using the following steps:

- Removed missing values in the target variable (Life expectancy)
- Encoded categorical variable (Status)
- Dropped high-cardinality column (Country)
- Handled missing values using mean imputation
- Split data into training (80%) and testing (20%) sets

Machine Learning Models

Three regression models were used:

- **Linear Regression** (baseline model)
- **Decision Tree** (captures non-linear relationships)
- **Random Forest** (ensemble model for better accuracy)

Model Evaluation Metrics

Models were evaluated using:

- **MAE (Mean Absolute Error):** Average prediction error
- **RMSE (Root Mean Squared Error):** Penalizes large errors
- **R^2 Score:** Measures how well the model explains variance

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Results Summary

Model	**MAE**	**RMSE**	**R^2 Score**
Linear Regression	4.29	6.08	0.57
Decision Tree	1.49	2.56	0.92
Random Forest	1.05	1.66	0.97

Random Forest performed best, showing strong predictive accuracy and capturing complex relationships in the data.

Feature Importance

The most important features influencing life expectancy were:

- HIV/AIDS
- Income composition of resources
- Adult Mortality
- Schooling
- Under-five deaths

HIV/AIDS was the most influential factor, showing a strong impact on life expectancy across countries.

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Conclusion

The **Random Forest model** was selected as the final model due to its high accuracy and low error rates.

Key findings show that life expectancy is strongly influenced by:

- **Disease prevalence (especially HIV/AIDS)**
- **Income and education levels**
- **Mortality rates**

The final model can be used to predict life expectancy and support global health analysis.